sLight User Guide

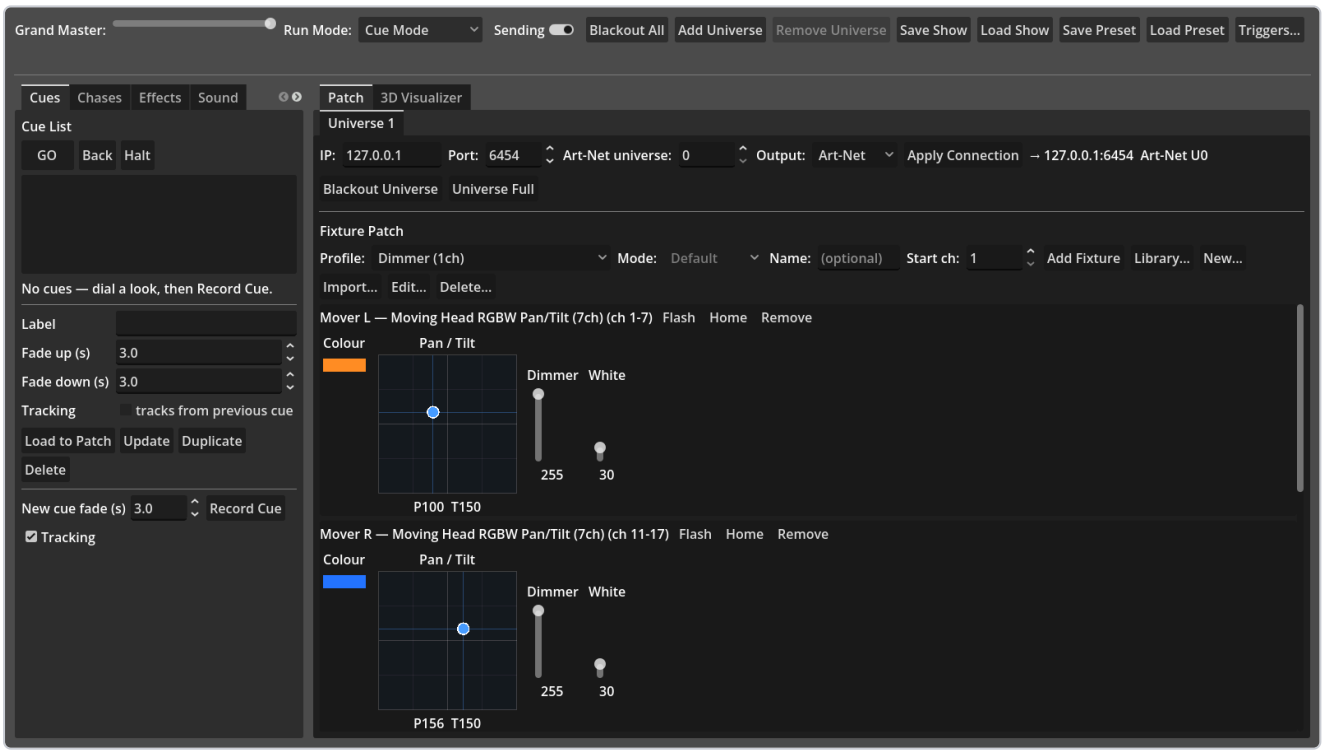
A step-by-step walkthrough for a new user, from opening the app for the first time to running a show. For a complete technical reference of every feature, see the README — this guide focuses on the essential workflow, in order, with screenshots.

What you need

sLight controls real (or simulated) DMX lighting fixtures. You don't need any hardware to follow this guide — see Testing without real hardware in the README for a free way to watch the DMX output live (QLC+ or an Art-Net monitor). When you're ready to control real lights, see Going to real fixtures.

Launch the sLight application — there's no separate setup step, no account, and no configuration file to edit first. The whole window you'll see below is what opens.

The window, at a glance



The Patch tab with two moving heads patched and lit

The window is split into three areas:

- **Top bar** — the **View** selector (**Console**, **DJ Edit Mode**, **DJ Perform Mode**), Grand Master fader, the **Run Mode** selector, **Sending** toggle, Add/Remove Universe, New/Save/Load Show and Preset, and **Triggers…** for MIDI/OSC. These are always visible, whatever tab or view you're on.
- **Left side — playback tabs**: **Cues**, **Chases**, **Effects**, **Sound**, **Groups**. This is where you build and run a show.

- **Right side — Patch / 3D Visualizer**: one tab per universe for patching fixtures and dialing them in by hand, and a 3D view of your rig lit live by whatever the console is currently outputting.

That split is the **Console** view. The top bar's **View** selector swaps it for **DJ Edit Mode** — where you build a light show for each song in your music library — or **DJ Perform Mode**, where VirtualDJ plays those shows live (Steps 10 and 11). The console keeps running underneath either one.

Everything below walks through these in the order you'd normally use them for the first time.

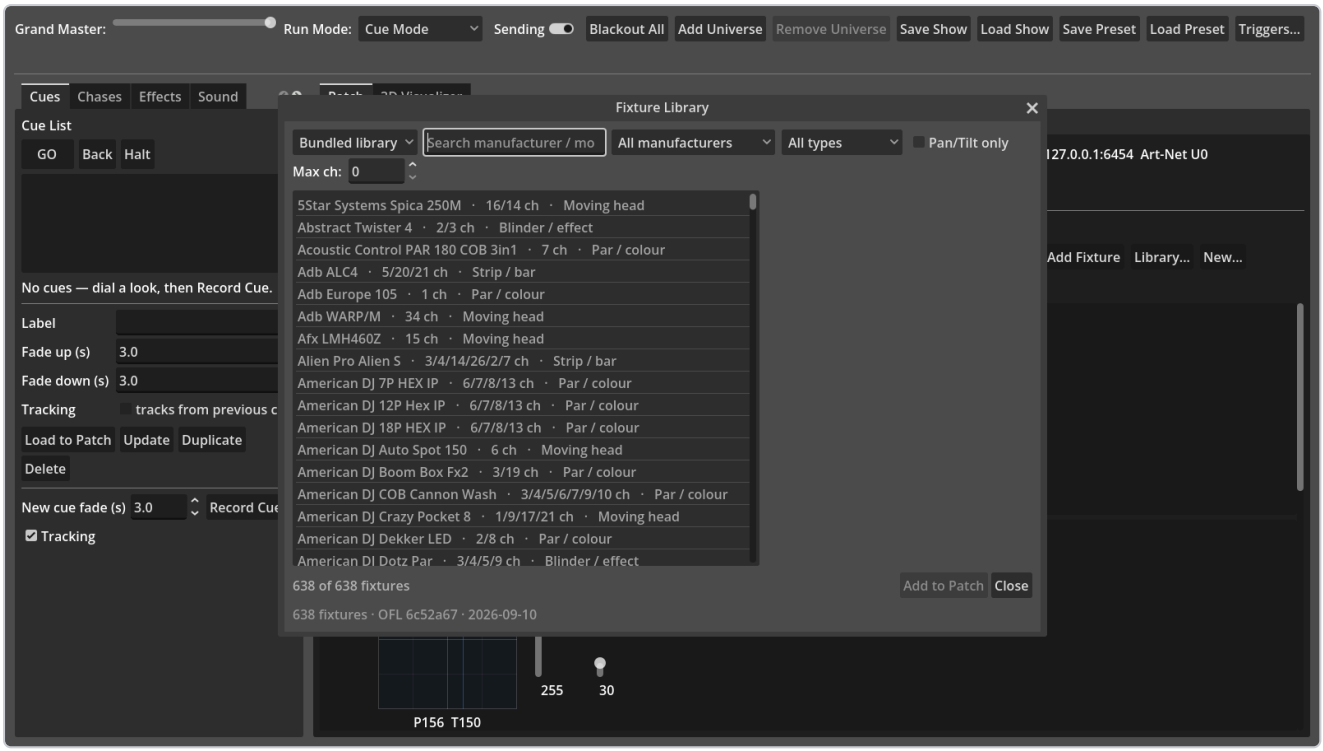
Step 1 — Add a universe and connect it

Each **Universe** tab (right side) is one 512-channel DMX universe. A fresh project starts with **Universe 1**. Pick an **Output** — Art-Net is the default and needs no extra hardware or plugins — set the target (for Art-Net, the IP/Port/universe number; `127.0.0.1:6454` universe `0` talks to a monitor running on the same machine), and click **Apply Connection**. Add more universes from **Add Universe** in the top bar (or **Ctrl+Shift+N**) if your rig needs more than 512 channels; **Remove Universe** (**Ctrl+Shift+R**) asks first, telling you how many fixtures go with it.

Step 2 — Patch your first fixture

"Patching" tells sLight what kind of light is plugged in where. In the **Fixture Patch** row, pick a **Profile** (a description of the fixture's DMX channels) and, if it has more than one, a **Mode**. Set the **Start ch:** to where it sits in the universe, optionally give it a **Name**, and click **Add Fixture**. Its control panel appears right below — a colour picker, a pan/tilt pad for moving heads, sliders for everything else (see the screenshot above).

A handful of generic profiles (Dimmer, RGB, RGBW, a moving head) ship built in, but real fixtures vary a lot — click **Library…** to search ~640 real fixtures converted from the Open Fixture Library:



Browsing the bundled fixture library

Search by manufacturer or model, filter by type/channel count/pan-tilt, preview a fixture's modes and channel layout, then **Add to Patch**. If you have a GDTF or Open Fixture Library file for something not in the bundle, **Import…** reads it directly.

Step 3 — Control a fixture by hand

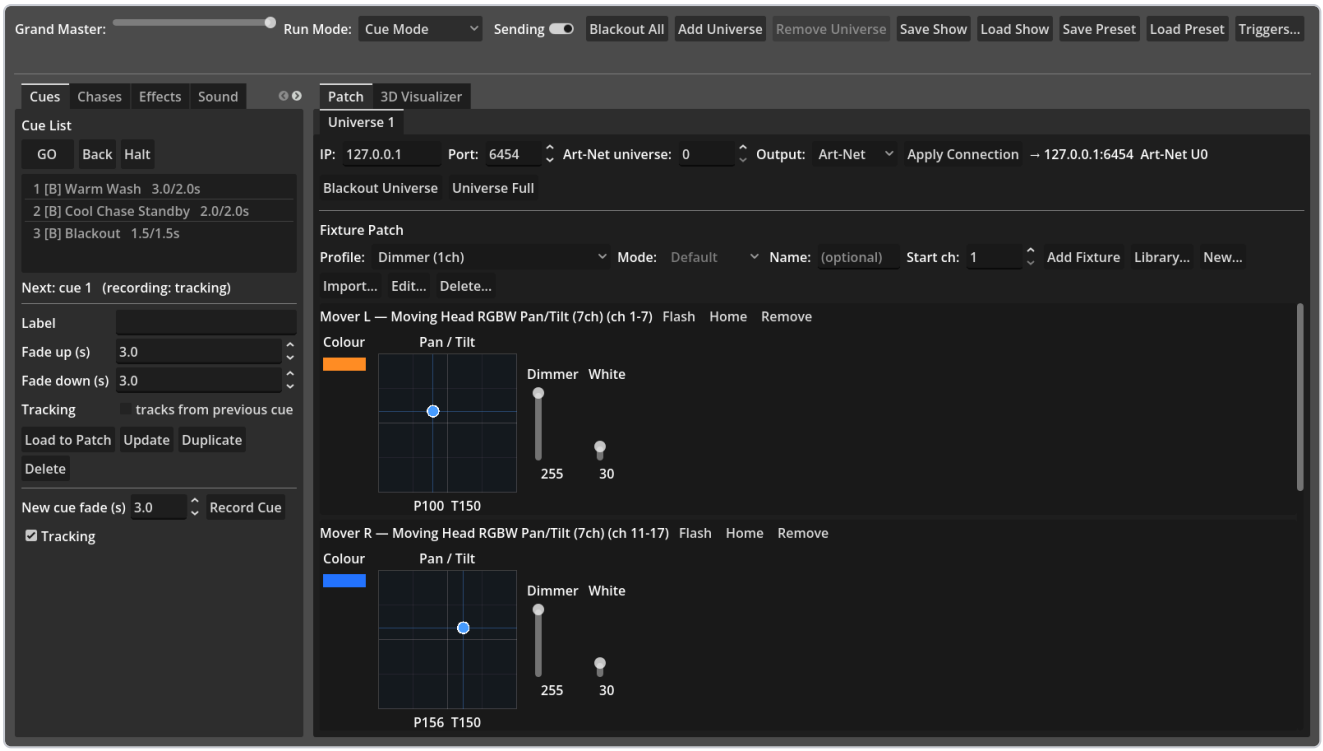
Every patched fixture gets purpose-built controls (visible in the screenshot for Step 2):

- A full Red/Green/Blue trio becomes one **colour picker**.
- A **Pan** + **Tilt** pair becomes a 2D pad — drag the puck to aim the head.
- Everything else gets its own slider, clamped to its usable range.

Flash (hold) snaps that fixture to full white and restores it exactly on release — handy for finding a fixture in a dark room. **Home** resets every control to its default. Fixture controls send live DMX output immediately (roughly 30 times a second, while **Sending** is on) — there's no separate "apply" step.

Step 4 — Record your first cues

Once you've dialed in a look you like, switch to the **Cues** tab and click **Record Cue** — it captures the current output of every universe as a new cue. Repeat for a few different looks, then step through them with **GO** (or just press the spacebar) — the whole rig crossfades to the next cue's look over its fade time.

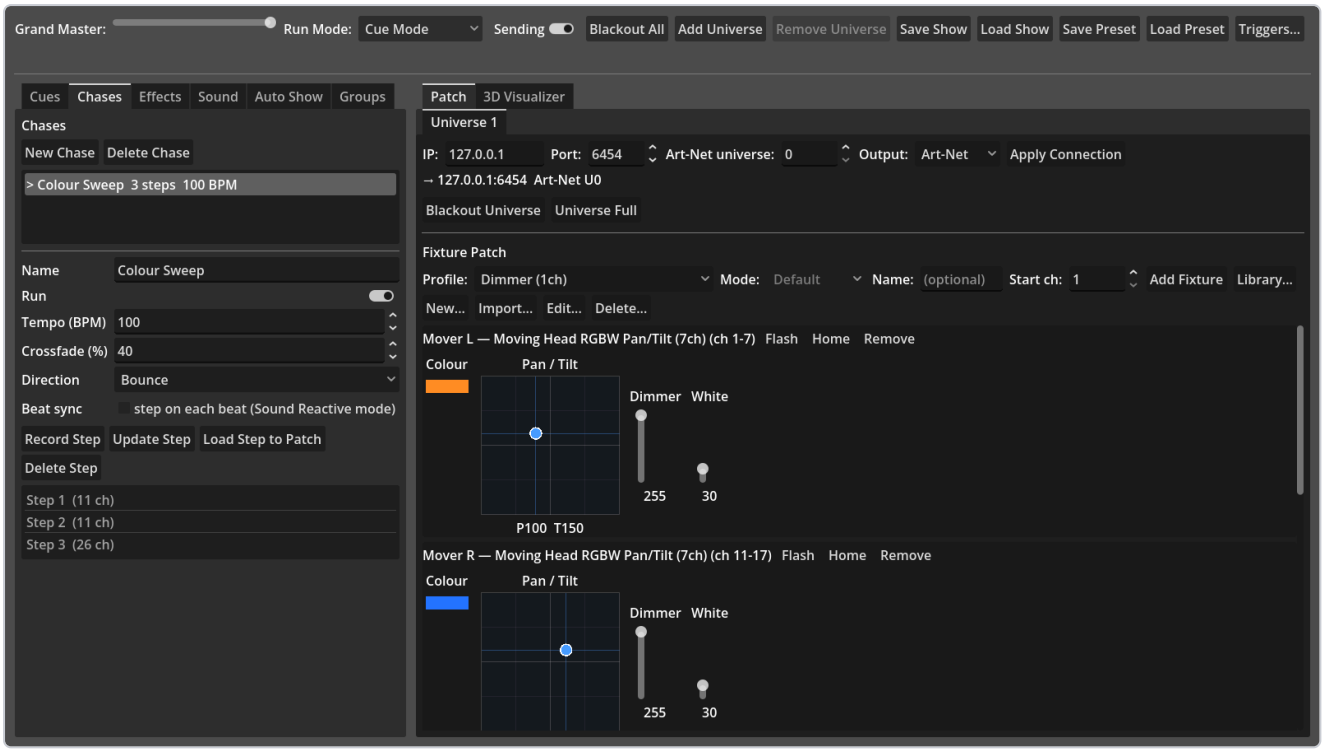


The Cues tab with three recorded cues

Select a cue to edit its **label** and **fade up/down** times inline. **Load to Patch** pulls a cue's stored look back into the fixture controls so you can tweak it and **Update** to re-record. See the README's Cue list section for the tracking vs. block distinction, which matters once you're editing a list of cues that build on each other.

Step 5 — Add movement with a chase

A **chase** is a list of captured steps that cycle automatically at a tempo — good for a repeating colour or position pattern that would be tedious to step through by hand. On the **Chases** tab, click **New Chase**, dial a look, click **Record Step**, and repeat for each step in the pattern.

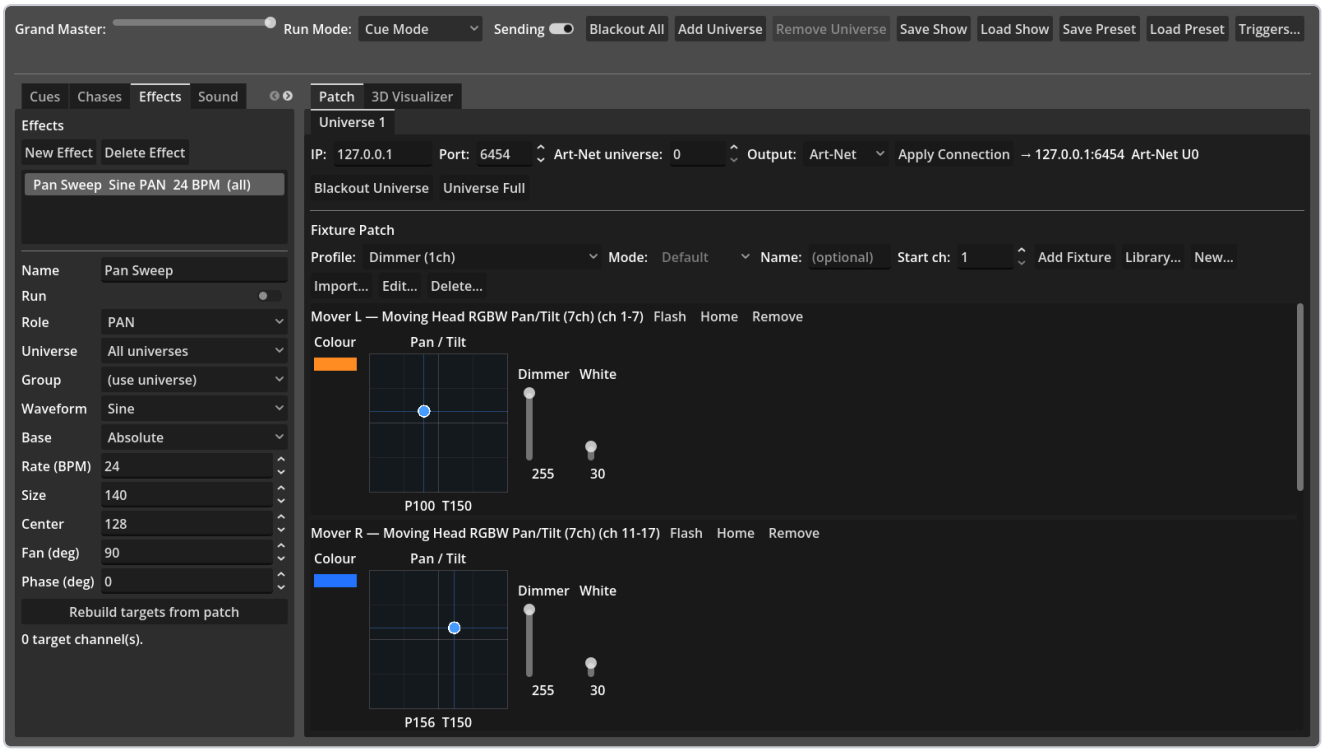


The Chases tab with a running 3-step colour chase

Set the **Tempo (BPM)**, how much each step **Crossfade**s in from the last one, and a **Direction** (forward / backward / bounce). Click **Run** to start it — a chase runs as a layer *on top of* whatever your cues are doing, so it doesn't disturb your programmed look, and stops cleanly when you turn it off.

Step 6 — Layer in an effect

An **effect** is a smooth waveform (sine, triangle, sawtooth, square, or random) applied continuously to one channel role — a breathing dimmer, a Pan/Tilt circle, a slow colour swell — without needing to record any steps.

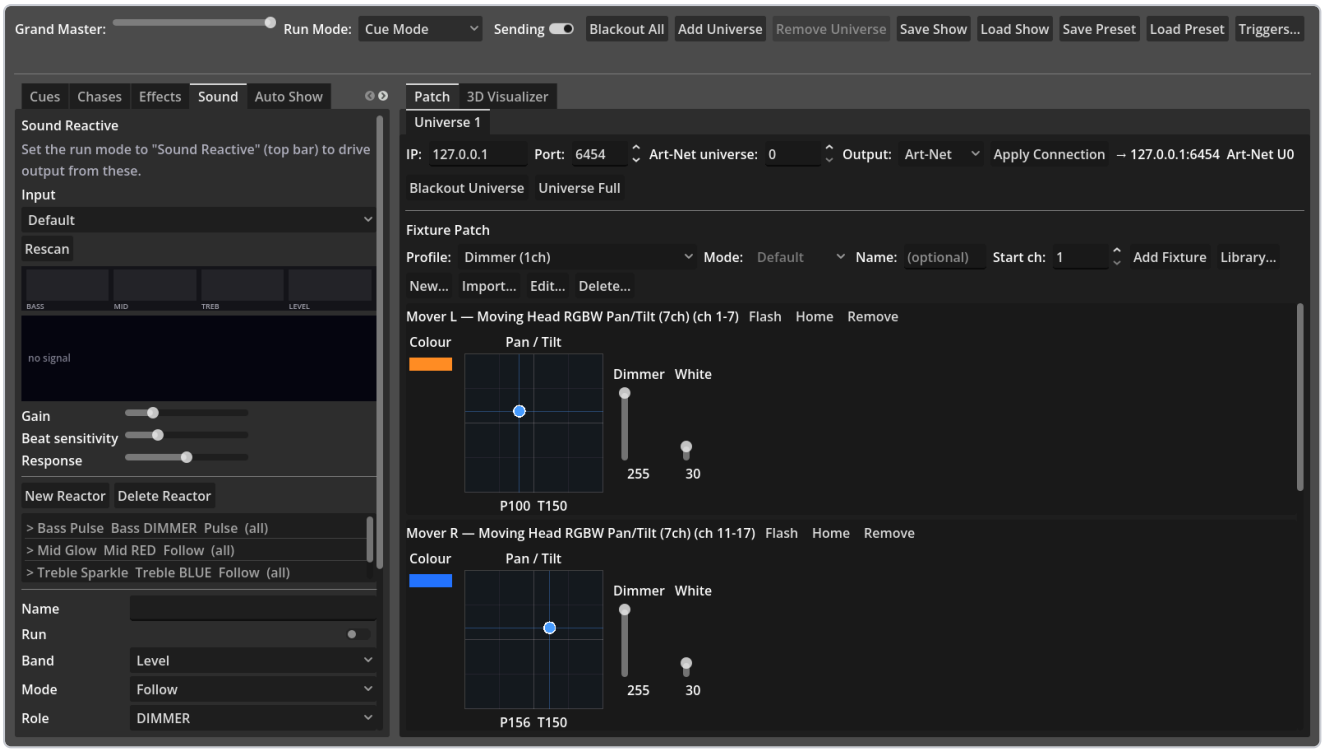


The Effects tab with a Pan Sweep effect configured

Pick the **Role** it drives and which fixtures (a **Universe**, or a **Group** — see Step 8), a **Waveform**, **Rate (BPM)**, and **Size** (how far it swings). **Fan (deg)** spreads the phase across the fixture list, so instead of every fixture moving in lockstep you get a chase-like ripple across the rig — try it with two Pan/Tilt effects 90° out of phase for a circle. Click **Run** to start it; like chases, effects layer over your cues without disturbing them.

Step 7 — React to music live

The top bar's **Run Mode** has two settings: **Cue Mode** (the default — the cue list drives playback) and **Sound Reactive**. Set it to **Sound Reactive** and switch to the **Sound** tab to drive output straight from a microphone or line input.

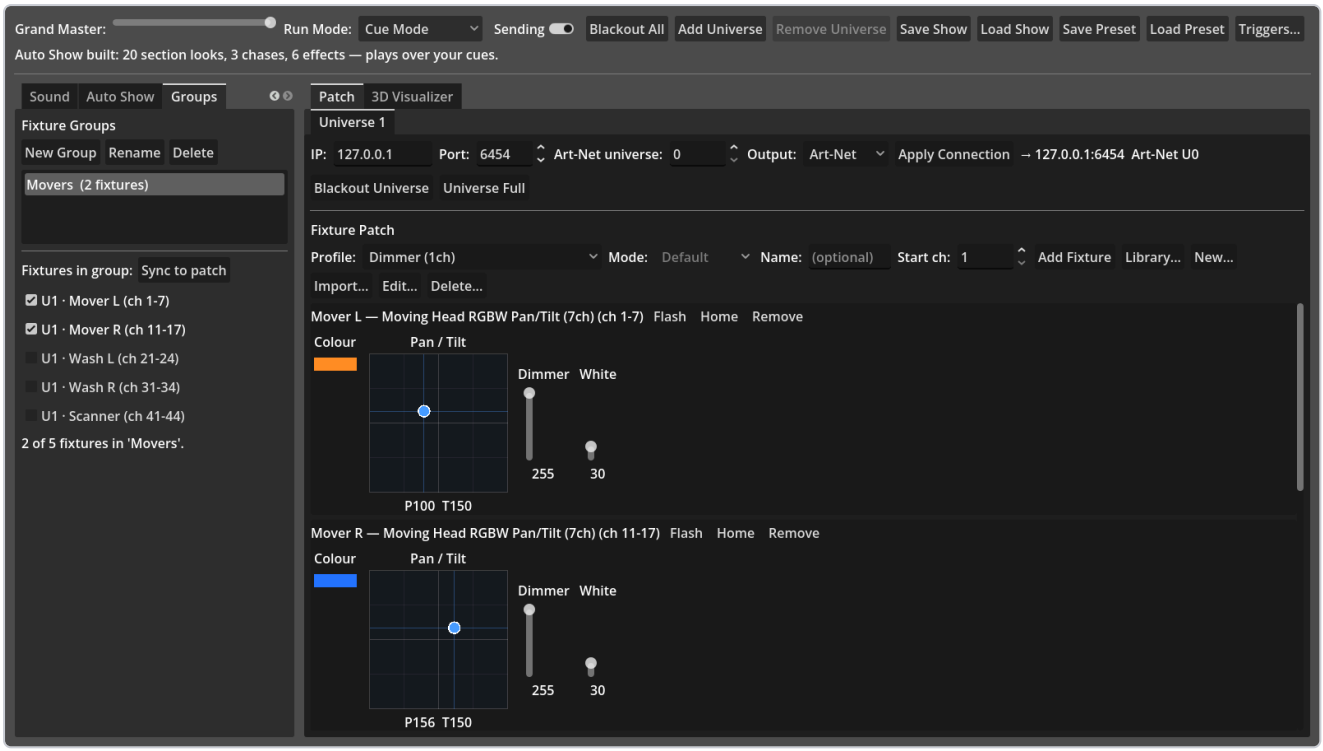


The Sound tab with the default starter reactors

Pick an **Input** device, and a live **bass / mid / treble / level** meter with a spectrogram shows what's coming in. A **reactor** maps one band (or the beat) onto a channel role — **Follow** mode tracks the band continuously (bass into a dimmer swell), **Pulse** mode snaps to full on every detected beat and releases (a beat-synced flash). A starter set of four reactors is already here to try — **Bass Pulse**, **Mid Glow**, **Treble Sparkle**, and **Beat Strobe** (off by default, since it's a lot more intense than the others) — edit or delete them, or add your own with **New Reactor**. Tick **Run** on the ones you want active.

Step 8 — Organize fixtures into groups

The **Groups** tab holds named sets of patched fixtures, so an effect or reactor can target just "the movers" or "the front truss" instead of a whole universe.

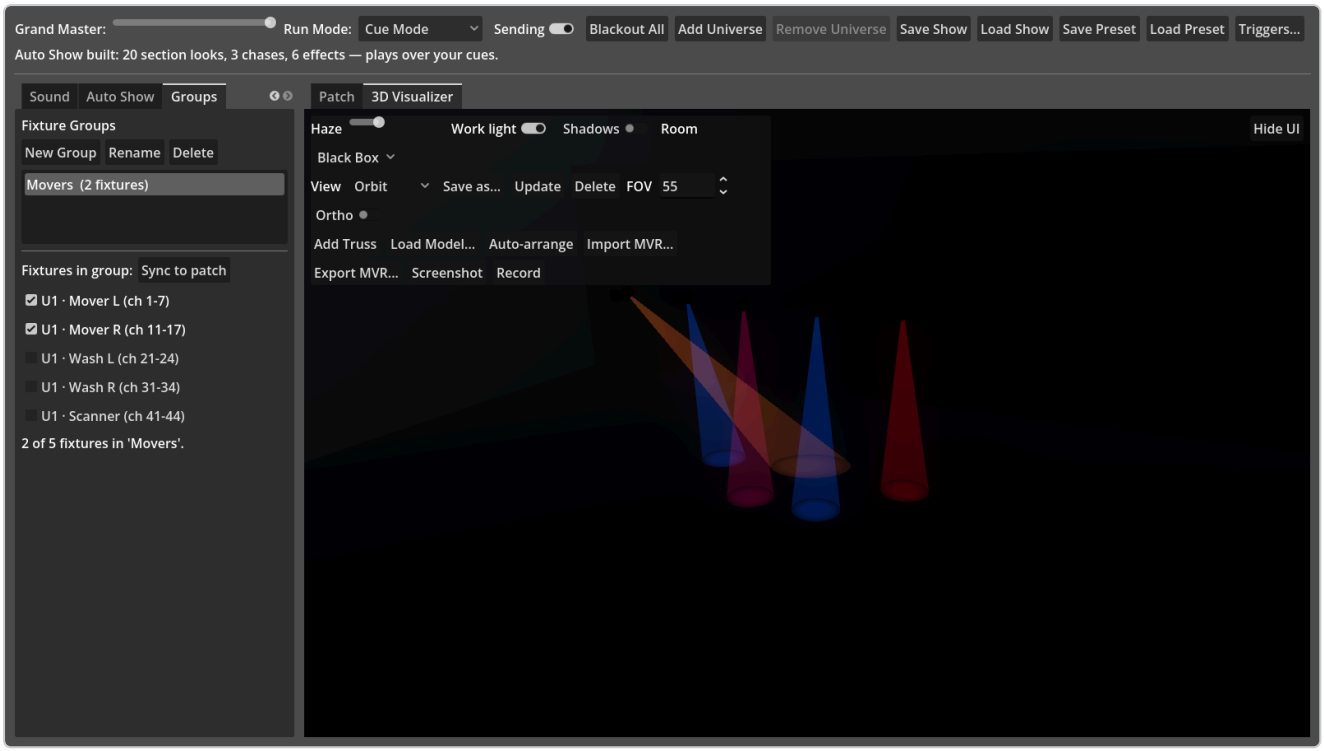


The Groups tab with a two-fixture "Movers" group

New Group, then tick which patched fixtures belong to it. Pick that group's name in an effect's or reactor's **Group** field instead of a universe filter. **Sync to patch** refreshes the checklist after you add or remove fixtures.

Step 9 — See it in 3D

The **3D Visualizer** tab (right side, next to Patch) shows your rig lit live by whatever the console is currently outputting — cues, chases, effects, and the grand master all show, updating continuously.



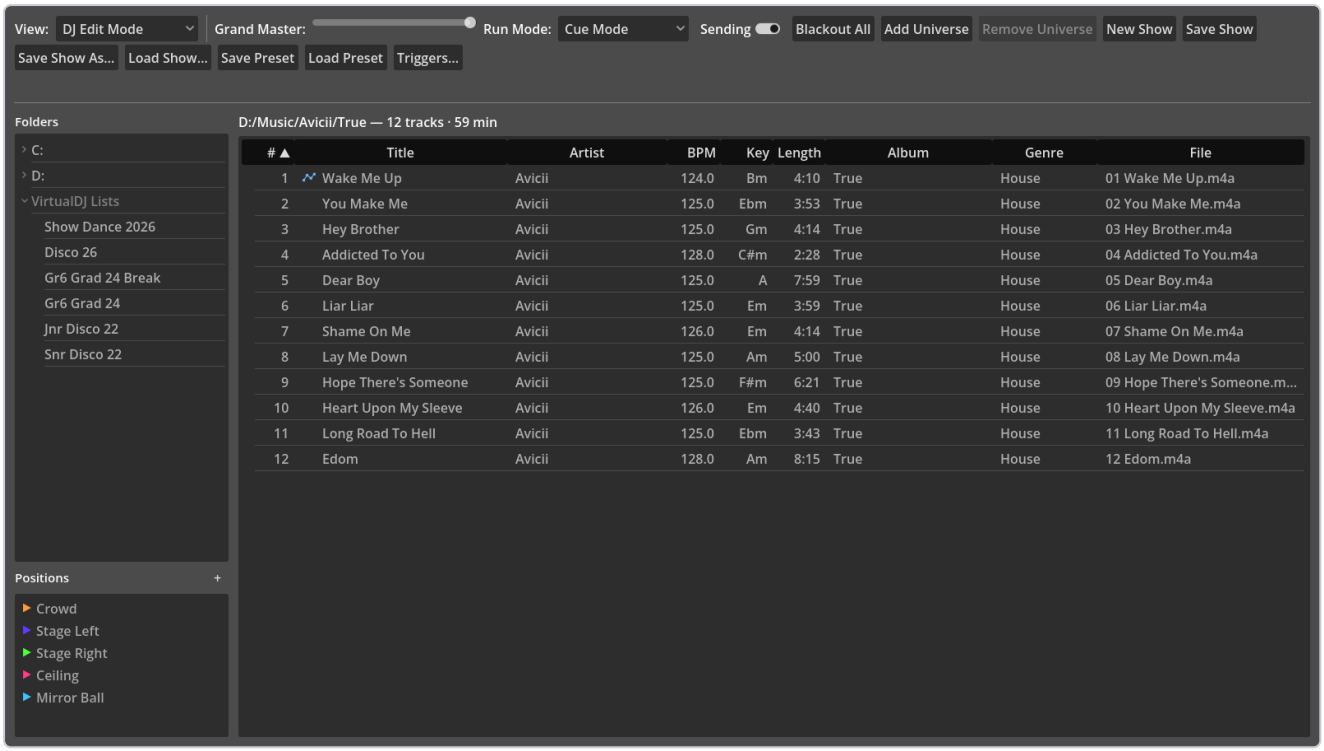
The 3D Visualizer showing a lit rig

Left-drag empty space to orbit the camera, wheel to zoom. **Auto-arrange** hangs every patched fixture in a tidy grid to start from; drag a fixture on the floor (or type exact X/Y/Z and heading/tilt) to place it precisely. **Add Truss** and **Load Model…** bring in rigging and set pieces. **Haze** adds real volumetric fog so beams show up in the air; **Room** switches between a few preset spaces. **Screenshot** and **Record** save a still or an MP4 straight from this view. Drag the **3D Visualizer** tab off the tab bar to pop it into its own window — handy for a second monitor at front-of-house.

Step 10 — Build a light show for a song (DJ Edit Mode)

Switch the **View** to **DJ Edit Mode** to program lighting against individual songs rather than cue by cue.

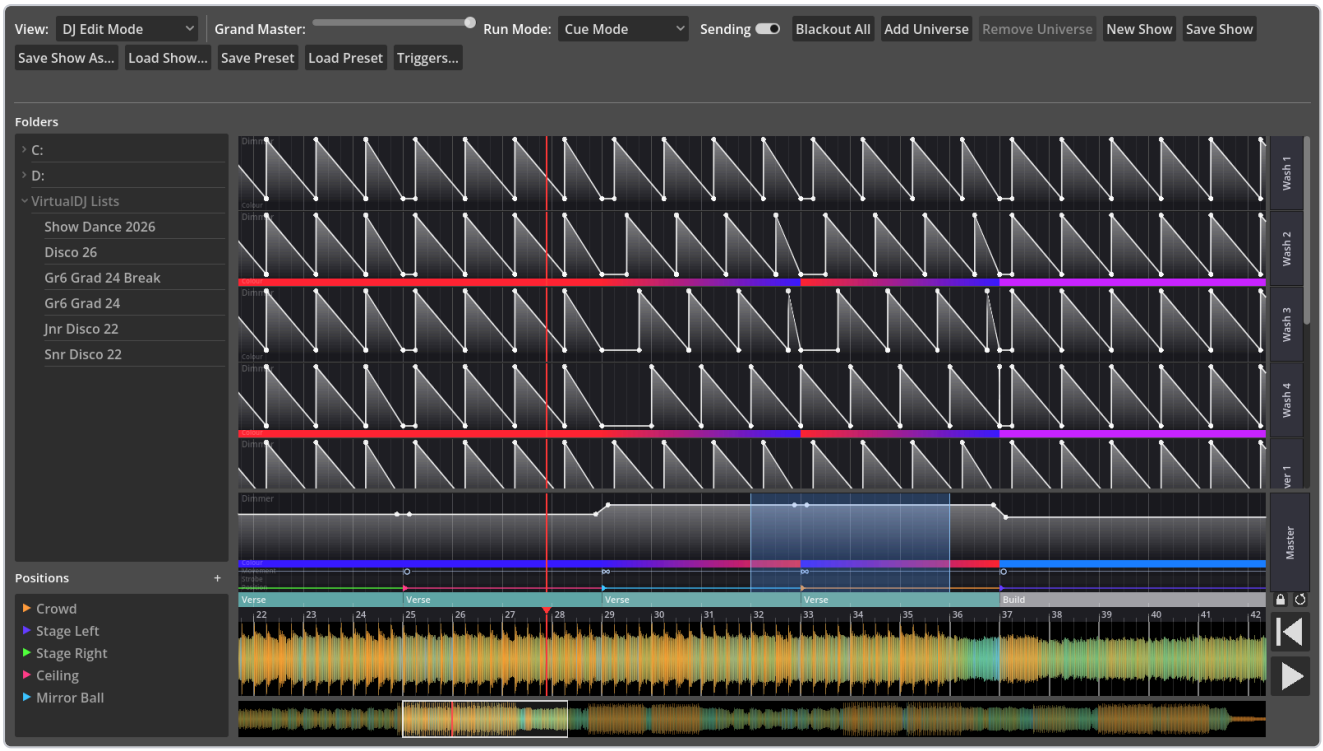
Find a song. On the left is a tree of this computer's drives and folders, followed by **VirtualDJ Lists** (your VirtualDJ Virtual Folders). Select one and its music is listed on the right — title, artist, BPM, key, length, album, genre — with details from VirtualDJ's database where it knows the track. Click a column title to sort. Songs that already have a light show in the current show are marked with a small blue node-line icon before the title. M4A / AAC songs open on Windows and macOS.



DJ Edit Mode listing an album, with a light show already on its first track

Open its timeline. Double-click a song. The list is replaced by its timeline (select a folder to go back). It opens fully zoomed out:

- Along the bottom is a colour waveform of the song, with VirtualDJ's beat grid drawn behind it and the bars numbered along the top. The mouse wheel zooms, **Shift**+wheel scrolls, and dragging with the middle mouse button pans. The slim overview beneath shows the whole song — drag its rectangle, or click to jump.
- **Play / Pause** (or **Space**) and **Back to start** sit to the right. The red playhead runs down every timeline; click or drag it on the waveform, the bar ruler or any timeline to move it.
- Above the waveform is the song's **structure** — Intro, Verse, Chorus and so on — worked out in the background when the song opens. It's locked by default; click the padlock to fix a section border the analyser put slightly wrong (drag it), or right-click a section to relabel, split or delete it. Right-click > **Snapping** chooses whether borders and selections snap to bars, beats or nothing.
- Above the structure is the **master** timeline, which drives every fixture, and above that a scrolling timeline for each patched fixture (and each head of a multi-head fixture).



A song's timelines after Generate, with four bars of the master selected

While the song plays, the lights and the 3D Visualizer follow the playhead exactly as they will in a live show — edits included. When it's paused, the look at the playhead holds, so you can scrub through and see each moment.

Let sLight start it for you. Click **Generate** (beside the padlock) to build a complete show from the song's structure and energy: the master's dimmer follows the music section by section, fixtures pulse on the beat in the loudest parts, colours change with each kind of section, a strobe builds before big rises in energy, and moving heads travel between your named positions with movement that grows with the energy. It replaces what the timelines had, as a single undo step — a good starting point to edit by hand.

Each timeline has rows for what its fixture can do:

- **Dimmer** — a line of nodes. Drag a node to move it (hold **Shift** to lock it to one axis), drag the line to raise or lower a stretch, **Ctrl**+click the line to add a node and **Ctrl**+right-click a node to remove it. The master's line starts at 0%, so nothing lights until you raise it. A fixture's (or head's) dimmer row starts empty and follows the master; give it values and it overrides the master over that stretch.
- **Colour** — segments of one static colour or a two-colour gradient.
- **Strobe** — stretches where the strobe runs at Slow, Medium or Fast.
- **Position** (moving heads) — which named position the heads point at (see *Named positions* below).
- **Movement** (moving heads) — a Circle, Figure 8, Pan Sweep or Tilt Sweep around that position, Small / Medium / Large, cycling every ½, 1 or 2 bars — or None to hold still.

Select, then edit. Drag across empty parts of the timelines to select an area — over every row the drag crosses, snapped to bars or beats. **Ctrl**-drag adds another area, **Shift**-drag extends the last one. **Right-click** the selection for a menu of what fits: Dimmer shapes (Black, Half, Full, Fade In / Out, Pulse In / Out, Smooth, Pulse on Every Beat / Bar), Colour (Static Colour…, Gradient…, Rainbow, presets), Strobe speeds,

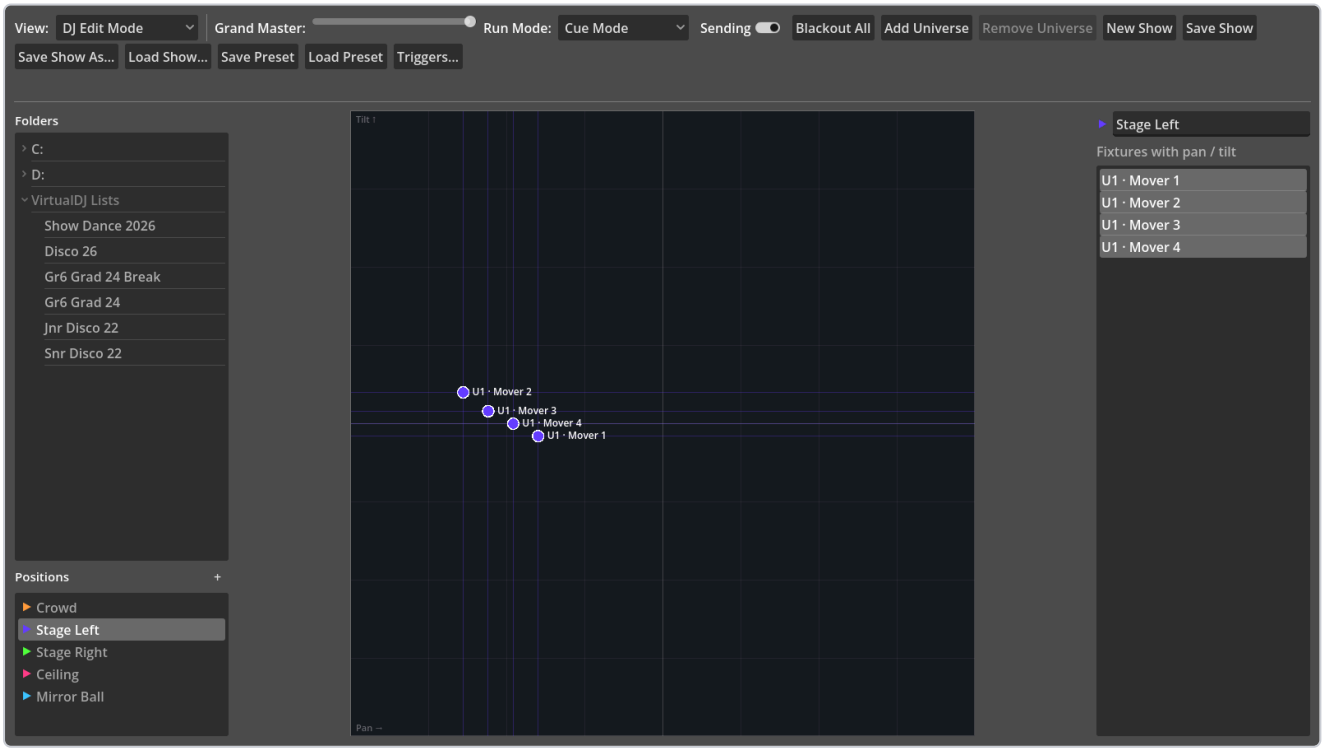
Position, Movement (with Size and Cycle), and Clear for each. Drag a selection's start or end to resize it (what sits on that edge moves with it), or drag its inside to move its content elsewhere — a ghost shows where it will land.

Keys	Does
Delete	Clear the selected areas
Ctrl+C / Ctrl+X / Ctrl+V	Copy / cut / paste — into every selected fixture at once, or at the playhead with nothing selected
Ctrl+Z / Ctrl+Y	Undo / redo (each song keeps its own history, up to 100 steps)
Ctrl+A	Select everything
Escape or **Ctrl+D**	Clear the selection

A song's timelines are saved automatically when you leave it — open another song, switch view, or close the app — and with **Save Show**. They're stored with the show (see *Saving your work*), keyed so a song that's been moved or renamed keeps its light show.

Named positions

Moving heads point at **named positions** rather than raw pan/tilt values. In DJ Edit Mode's left-hand tree, below the folders, the **Positions** header lists them; click its **+** to add one. It opens in the **position editor**: a large pan/tilt pad beside a list of every moving fixture. Select fixtures in the list (**Ctrl** / **Shift** for several) and their points light up on the pad in the position's colour — drag one to aim them all. While selected, those fixtures are lit full white and pointed live at the position, so you can see exactly where they land in the room; they go back to how they were when you leave. Rename the position at the top of the editor, and right-click one in the list to **Rename**, **Duplicate** or **Delete** it.

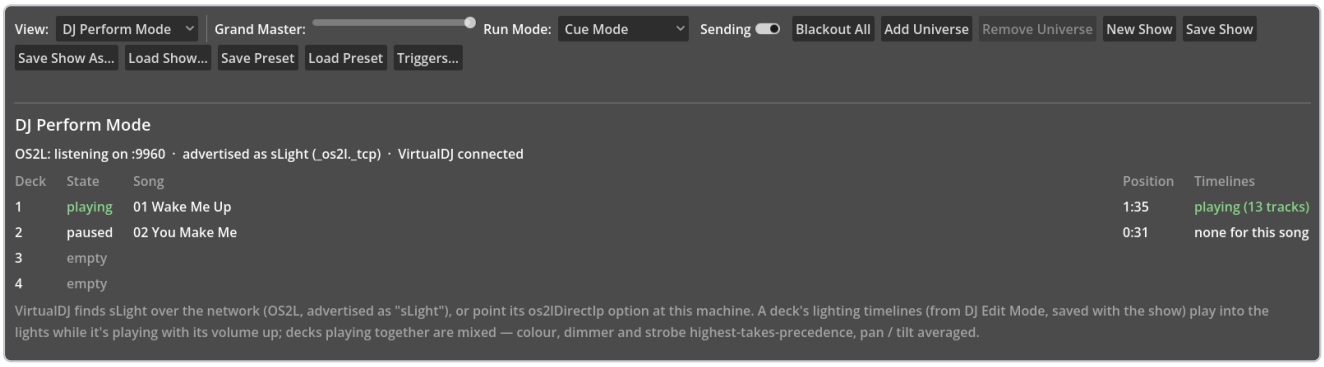


The position editor with four movers aimed at "Stage Left"

Use them in a timeline's Position row (right-click > **Position**), or let **Generate** spread them across the song's sections.

Step 11 — Play it live with VirtualDJ (DJ Perform Mode)

Switch the **View** to **DJ Perform Mode** and sLight starts listening for VirtualDJ over OS2L. It's advertised on the network as "sLight" (port 9960), so VirtualDJ normally finds it on its own; otherwise point VirtualDJ's `os2lDirectIp` setting at this computer. The panel shows the connection and, for each of up to four decks, the song loaded, whether it's playing, its position and whether its timelines have loaded.



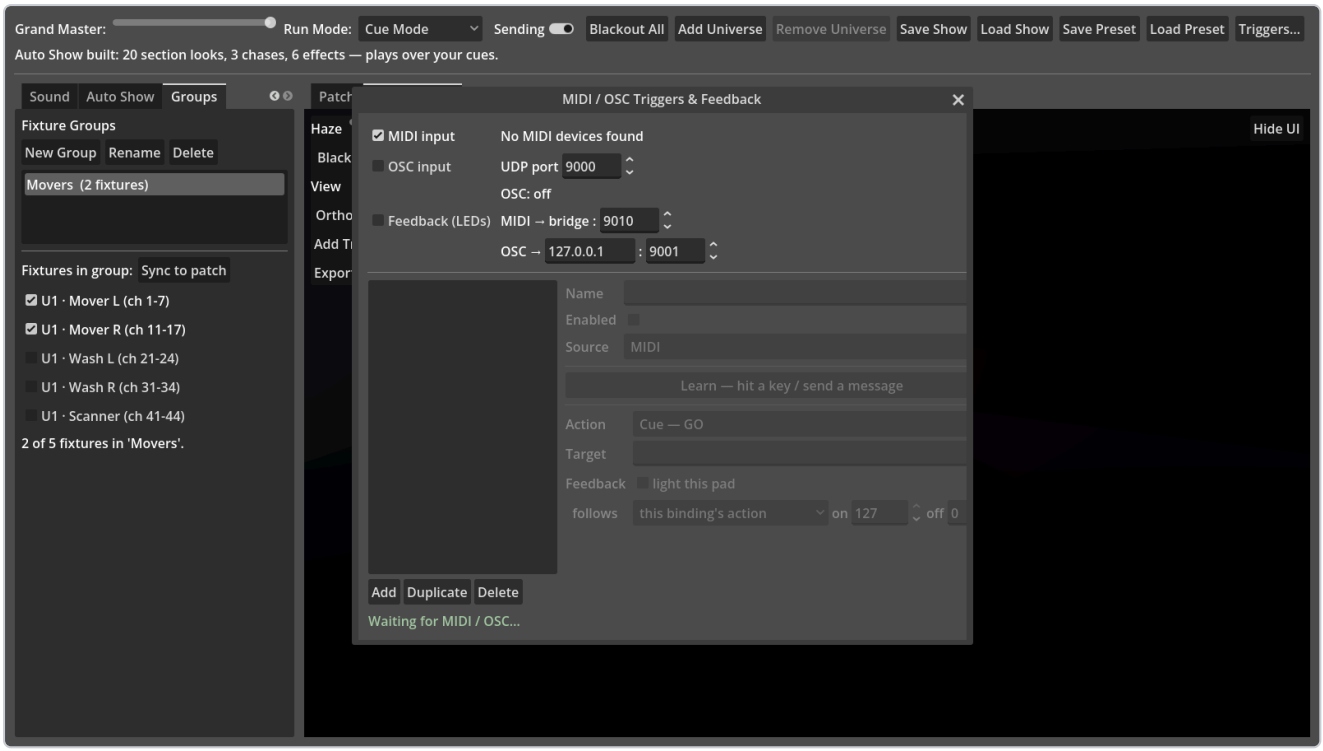
DJ Perform Mode with VirtualDJ connected and one deck playing its light show

Whenever a deck loads a song, its light show from DJ Edit Mode loads in the background. A deck drives the lights while it's **playing with its volume fader up** — a track you're cueing in your headphones with the fader down stays dark — following VirtualDJ's position and pitch. When two decks play together during a mix, their shows blend: brightest wins for colour, dimmer and strobe, and pan/tilt are averaged.

Only what a song's timelines cover is driven; everything else stays as the console has it, and chases and effects still layer on top. Leaving DJ Perform Mode stops the listener and the playback.

Optional: MIDI and OSC control surfaces

Triggers… in the top bar (works from any tab) opens the binding editor, so a MIDI controller or an OSC app like TouchOSC can fire cues, toggle chases/effects, or trigger a blackout.



The MIDI/OSC Triggers dialog

Turn on **MIDI input** or **OSC input**, click **Add**, then **Learn** and play the pad / move the fader / send the message you want bound — the match fields fill in automatically. Pick an **Action** (Cue GO, a chase or effect toggle, blackout, …). **Feedback (LEDs)** can light a pad back based on what it controls, or as a standalone beat/sending/run-mode indicator. This is entirely optional — most of the app works fine with just a mouse and keyboard.

MIDI feedback needs somewhere to send it: the screenshot above shows the default **MIDI → bridge :** field, which needs a small helper script running (`python tools/midi_bridge.py` , see the README). If the optional `native` extension is built, this row becomes a **MIDI output** port picker instead — pick a real MIDI port directly, no extra script needed.

Saving your work

A show lives in one `.slight` file you choose, holding everything — every universe's connection settings and patch, cues, chases, effects, groups, named positions, sound reactors, MIDI/OSC bindings and the visualizer's stage. The songs' lighting timelines go in a folder beside it (`MyShow.slight` → `MyShow.timelines/`), so keep the two together when you copy a show to another computer.

Top bar	Keys	Does
Save Show	**Ctrl+S**	Save to the current show file (asks where, for a new show)
Save Show As…	**Shift+Ctrl+S**	Save to a new file, copying the timelines folder alongside
Load Show…	**Ctrl+O**	Open a show
New Show	**Ctrl+N**	After asking, start an empty show (your output settings are kept)

The show you last saved or loaded reopens automatically the next time you start sLight. **Save/Load Preset** is a lighter-weight snapshot of just the live DMX output, for a quick "go back to how it looked a minute ago" without touching your cue list.

Keyboard shortcuts

Keys	Does
Space	GO the next cue (Console, Cue Mode) / play or pause the song (DJ Edit Mode)
Ctrl+N / Ctrl+O	New show / load a show
Ctrl+S / Shift+Ctrl+S	Save show / save show as
Ctrl+Shift+N / Ctrl+Shift+R	Add a universe / remove the current one

DJ Edit Mode's timeline editing shortcuts are listed in Step 10.

Where to go next

This guide covers the everyday path from an empty patch to a running show. The README is the complete reference — fixture profile details (multi-mode personalities, 16-bit fine channels, named value ranges for gobo/colour wheels), sACN and USB DMX output, cue tracking vs. blocks, MVR import/export, every run mode's exact behaviour, and exactly how DJ Perform Mode mixes decks and drives strobe, position and movement channels are all there if you want to go deeper.